

Qu. 1.

- a) Find the equation of the tangent plane to the surface

$$z = x^3 - y^3 + \sin(xy)$$

at the point $(0, -1, 1)$.

- b) Let $f(x, y, z) = xyz + x^2y^3$. Find the directional derivative of f at the point $(0, -2, 3)$ in the direction to the point $(4, 1, 2)$.

Solution.

- a) $\nabla f(a, b, c)$ is normal to the tangent plane to the level surface $f(x, y, z) = k$ at the point (a, b, c) . The vector equation of the tangent plane is then

$$\mathbf{r} \cdot \nabla f(a, b, c) = (a, b, c) \cdot \nabla f(a, b, c)$$

where $\mathbf{r}$ is the position vector of the point (x, y, z) .

Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ be the function $f(x, y, z) = x^3 - y^3 + \sin xy - z$, so that the surface given is the level surface $f(x, y, z) = 0$.

$\nabla f = (3x^2 + y \cos xy, -3y^2 + x \cos xy, -1)$, giving at $(0, -1, 1)$ the vector $(-1, -3, -1)$

So the equation of the plane is given by

$$(x, y, z) \cdot (-1, -3, -1) = (0, -1, 1) \cdot (-1, -3, -1)$$

or $x + 3y + z = -2$

- b) The gradient of f is given by

$$\nabla f = (yz + 2xy^3)\mathbf{i} + (xz + 3x^2y^2)\mathbf{j} + xy\mathbf{k}$$

and once evaluated at $(0, -2, 3)$ it becomes

$$\nabla f(0, -2, 3) = -6\mathbf{i}.$$

A unit vector pointing from $(0, -2, 3)$ to $(4, 1, 2)$ is

$$\vec{u} = \frac{4\mathbf{i} + 3\mathbf{j} - \mathbf{k}}{\sqrt{26}}$$

The directional derivative $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$.

Finally, we have

$$D_{\mathbf{u}}f(0, -2, 3) = \nabla f(0, -2, 3) \cdot \mathbf{u} = -6\mathbf{i} \cdot \frac{4\mathbf{i} + 3\mathbf{j} - \mathbf{k}}{\sqrt{26}} = -\frac{24}{\sqrt{26}}.$$

Qu. 2. For each of the following functions find (i) ∇f and (ii) the directional derivative in the direction of $\mathbf{u}$:

- a) $f(x, y) = y \sin x$ where $\mathbf{u} = 2\mathbf{i} - \mathbf{j}$;
- b) $f(x, y, z) = xe^{yz}$ at the point $(\pi, 1, -2)$ where $\mathbf{u} = \mathbf{i} - \mathbf{j} + 3\mathbf{k}$;
- c) $f(x, y) = x^2y + y^3x$ at the point $(1, 2)$ where $\mathbf{u} = (3, 4)$.

Solution. Let $\hat{\mathbf{u}} = \mathbf{u}/|\mathbf{u}|$ be the unit vector in direction $\mathbf{u}$, then $\nabla f(x, y) = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j}$

and $D_{\hat{\mathbf{u}}}f(x, y) = \nabla f(x, y) \cdot \hat{\mathbf{u}}$.

a) i) $\nabla f(x, y) = y \cos x \mathbf{i} + \sin x \mathbf{j}$.

ii) Again $|\mathbf{u}| = \sqrt{5}$, so $\hat{\mathbf{u}} = 2\mathbf{i}/\sqrt{5} - \mathbf{j}/\sqrt{5}$ and

$D_{\hat{\mathbf{u}}}f(x, y) = \nabla f(x, y) \cdot \hat{\mathbf{u}} = 2y \cos x/\sqrt{5} - \sin x/\sqrt{5}$.

b) i) $\nabla f(x, y, z) = e^{yz}\mathbf{i} + xze^{yz}\mathbf{j} + xye^{yz}\mathbf{k}$ so $\nabla f(\pi, 1, -2) = e^{-2}(\mathbf{i} - 2\pi\mathbf{j} + \pi\mathbf{k})$.

ii) $|\mathbf{u}| = \sqrt{1^2 + (-1)^2 + 3^2} = \sqrt{11}$, $\hat{\mathbf{u}} = (\mathbf{i} - \mathbf{j} + 3\mathbf{k})/\sqrt{11}$ and

$D_{\hat{\mathbf{u}}}f(\pi, 1, -2) = \nabla f(\pi, 1, -2) \cdot \hat{\mathbf{u}} = e^{-2}(\mathbf{i} - 2\pi\mathbf{j} + \pi\mathbf{k}) \cdot (\mathbf{i} - \mathbf{j} + 3\mathbf{k})/\sqrt{11} = e^{-2}(1 + 5\pi)/\sqrt{11}$.

c) A unit vector in the direction of $(3, 4)$ is $\mathbf{u} = (\frac{3}{5}, \frac{4}{5})$. The directional derivative $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$.

i) $\nabla f = (2xy + y^3, x^2 + 3y^2x)$ or $(12, 13)$ at $(1, 2)$.

ii) $D_{\mathbf{u}}(1, 2) = (12, 13) \cdot (\frac{3}{5}, \frac{4}{5}) = 88/5$.

Qu. 3. Find the directional derivative of the function $f(x, y, z) = (y^2 + \sin z)e^{-x}$ at $(1, 2, \pi/2)$ in the direction that points towards $(1, 1, 1)$.

Solution. A vector pointing towards $(1, 1, 1)$ from $(1, 2, \pi/2)$ is $(1-1)\mathbf{i} + (1-2)\mathbf{j} + (1-\pi/2)\mathbf{k} = -\mathbf{j} + (1-\pi/2)\mathbf{k}$ and so

$$\mathbf{u} = \frac{1}{\sqrt{\pi^2/4 - \pi + 2}}[-\mathbf{j} + (1 - \pi/2)\mathbf{k}]$$

is a unit vector in this direction. We want $D_{\mathbf{u}}(1, 2, \pi/2) = \nabla f(1, 2, \pi/2) \cdot \mathbf{u}$, but $\nabla f(x, y, z) = -(y^2 + \sin z)e^{-x}\mathbf{i} + 2ye^{-x}\mathbf{j} + e^{-x}\cos z\mathbf{k}$ so $\nabla f(1, 2, \pi/2) = -5e^{-1}\mathbf{i} + 4e^{-1}\mathbf{j}$.

Hence $D_{\mathbf{u}}(1, 2, \pi/2) = \nabla f(1, 2, \pi/2) \cdot \mathbf{u} = \frac{-4}{e\sqrt{\pi^2/4 - \pi + 2}}$.

Qu. 4. Find the equations of the tangent planes at the given points:

- a) $4z = x^2 + 4y^2$ at the point $(2, 1, 2)$;
- b) $xy^2 + 3x - z^2 = 4$ at the point $(2, 1, -2)$;
- c) $Ax^2 + By^2 + Cz^2 = K$ at the point (x_1, y_1, z_1) ;
- d) $z = e^{xy^2}$ at the point $(1, 1, e)$.

Solution. Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ be a function of the variables x, y and z , then the vector $\nabla f(a, b, c)$ is normal to the level surface $f(x, y, z) = k$ passing through (a, b, c) (so that $k = f(a, b, c)$). Hence $\nabla f(a, b, c)$ is normal to the tangent plane to the level

surface $f(x, y, z) = k$ at the point (a, b, c) . The Cartesian equation of the tangent plane to the surface $f(x, y, z) = k$ can now be calculated from the vector equation

$$\mathbf{r} \cdot \nabla f(a, b, c) = (a, b, c) \cdot \nabla f(a, b, c)$$

where $\mathbf{r}$ is the position vector of the point (x, y, z) .

(a) Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ be the function $f(x, y, z) = x^2 + 4y^2 - 4z$, so that the surface given is the level surface $f(x, y, z) = 0$. Now

$$\nabla f(x, y, z) = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k} = 2x\mathbf{i} + 8y\mathbf{j} - 4\mathbf{k}$$

so
$$\nabla f(2, 1, 2) = 4\mathbf{i} + 8\mathbf{j} - 4\mathbf{k}$$

Hence the tangent plane has equation

$$(x, y, z) \cdot (4\mathbf{i} + 8\mathbf{j} - 4\mathbf{k}) = (2, 1, 2) \cdot (4\mathbf{i} + 8\mathbf{j} - 4\mathbf{k})$$

which gives
$$4x + 8y - 4z = 8 + 8 - 8$$

or
$$x + 2y - z = 2$$

(b) Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ be the function $f(x, y, z) = xy^2 + 3x - z^2 - 4$, so that the surface given is the level surface $f(x, y, z) = 0$. Now

$$\nabla f(x, y, z) = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k} = (y^2 + 3)\mathbf{i} + 2xy\mathbf{j} - 2z\mathbf{k}$$

so
$$\nabla f(2, 1, -2) = 4\mathbf{i} + 4\mathbf{j} + 4\mathbf{k}$$

Hence the tangent plane has equation

$$(x, y, z) \cdot (4\mathbf{i} + 4\mathbf{j} + 4\mathbf{k}) = (2, 1, -2) \cdot (4\mathbf{i} + 4\mathbf{j} + 4\mathbf{k})$$

which gives
$$4x + 4y + 4z = 8 + 4 - 8$$

or
$$x + y + z = 1$$

(c) Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ be the function $f(x, y, z) = Ax^2 + By^2 + Cz^2 - K$, so that the surface given is the level surface $f(x, y, z) = 0$. Now

$$\nabla f(x, y, z) = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k} = 2Ax\mathbf{i} + 2By\mathbf{j} + 2Cz\mathbf{k}$$

so
$$\nabla f(x_1, y_1, z_1) = 2Ax_1\mathbf{i} + 2By_1\mathbf{j} + 2Cz_1\mathbf{k}$$

Hence the tangent plane has equation

$$(x, y, z) \cdot (2Ax_1\mathbf{i} + 2By_1\mathbf{j} + 2Cz_1\mathbf{k}) = (2, 1, -2) \cdot (2Ax_1\mathbf{i} + 2By_1\mathbf{j} + 2Cz_1\mathbf{k})$$

which gives
$$2(Ax_1x + By_1y + Cz_1z) = 2(Ax_1^2 + Cy_1^2 + Bz_1^2)$$

or
$$Ax_1x + By_1y + Cz_1z = K$$

(as (x_1, y_1, z_1) lies on the surface, $f(x_1, y_1, z_1) = K$).

(d) Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ be the function $f(x, y, z) = e^{xy^2} - z$, so that the surface given is the level surface $f(x, y, z) = 0$.

$$\nabla f = (y^2 e^{xy^2}, 2xy e^{xy^2}, -1),$$
 giving $(e, 2e, -1)$ at $(1, 1, e)$.

So the equation of the plane is given by

$$(x, y, z) \cdot (e, 2e, -1) = (1, 1, e) \cdot (e, 2e, -1)$$

or $ex + 2ey - z = 2e$.

Qu. 5. Evaluate $\iint_R (x - y) dA$ as a repeated integral where R is the triangle with vertices at $(3, 10)$, $(3, 1)$, $(-2, 1)$. [Hint: first find the equations of the lines of the triangle.]

Solution. The line through $(-2, 1)$ and $(3, 10)$ has equation $(y - 10)/(x - 3) = (10 - 1)/(3 - (-2))$ or $y = \frac{9}{5}x + \frac{23}{5}$. Since we have an expression for y in terms of x , we integrate with respect to y first keeping x fixed between -2 and 3 . For such an x , y varies from 1 to $\frac{9}{5}x + \frac{23}{5}$, so the integral becomes

$$\begin{aligned} I &= \int \int_R (x - y) dA = \int_{x=-2}^{x=3} \int_{y=1}^{y=\frac{9}{5}x + \frac{23}{5}} (x - y) dy dx = \int_{x=-2}^{x=3} \left[xy - \frac{y^2}{2} \right]_{y=1}^{\frac{9}{5}x + \frac{23}{5}} dx \\ &= \int_{x=-2}^{x=3} \left(x \left(\frac{9}{5}x + \frac{23}{5} \right) - \frac{1}{2} \left(\frac{9}{5}x + \frac{23}{5} \right)^2 - x + \frac{1}{2} \right) dx \\ &= \int_{x=-2}^{x=3} \left(x \left(\frac{9}{5}x + \frac{18}{5} \right) - \frac{1}{2} \left(\frac{9}{5}x + \frac{23}{5} \right)^2 + \frac{1}{2} \right) dx \\ &= \left[\frac{9}{15}x^3 + \frac{18}{10}x^2 - \frac{1}{6} \cdot \frac{5}{9} \left(\frac{9}{5}x + \frac{23}{5} \right)^3 + \frac{1}{2}x \right]_{-2}^3 \\ &= \left[\frac{3}{5}x^3 + \frac{9}{5}x^2 - \frac{5}{54} \left(\frac{9}{5}x + \frac{23}{5} \right)^3 + \frac{1}{2}x \right]_{-2}^3 = -60 \end{aligned}$$

Alternatively, one could expand each line completely at each stage:

$$\begin{aligned} I &= \int_{x=-2}^{x=3} \int_{y=1}^{y=\frac{9}{5}x + \frac{23}{5}} (x - y) dy dx = \int_{x=-2}^{x=3} \left[xy - \frac{y^2}{2} \right]_{y=1}^{\frac{9}{5}x + \frac{23}{5}} dx \\ &= \int_{x=-2}^{x=3} \left(x \left(\frac{9}{5}x + \frac{23}{5} \right) - \frac{1}{2} \left(\frac{9}{5}x + \frac{23}{5} \right)^2 - x + \frac{1}{2} \right) dx \\ &= \int_{x=-2}^{x=3} \left(\frac{9}{50}x^2 - \frac{117}{25}x - \frac{252}{25} \right) dx = \left[\frac{3}{50}x^3 - \frac{117}{50}x^2 - \frac{252}{25}x \right]_{-2}^3 = -60 \end{aligned}$$

Note that for ‘most’ values of x and y on the triangle R , $y > x$, so we expect the answer to be negative. Note also that this question is *not* really hard at all, it just requires a bit of care and checking to get the algebra right. After all the worst it gets to is integrating a quadratic.

Qu. 6. By evaluating an appropriate double integral, find the area enclosed between the two parabolas $x = y^2$ and $x = 2y - y^2$.

Solution. Let R be the area enclosed between the two parabolas, so $R = \{(x, y) : x \geq y^2, x \leq 2y - y^2\}$. The area required is the given by $A = \iint_R 1 dA$. From the diagram it is clear that for fixed $0 \leq y \leq 1$, $y^2 \leq x \leq 2y - y^2$, so we can express R as $R = \{(x, y) : 0 \leq y \leq 1, y^2 \leq x \leq 2y - y^2\}$, so integrating with respect to x first, we have

$$\begin{aligned} A &= \iint_R 1 dA = \int_{y=0}^{y=1} \int_{x=y^2}^{x=2y-y^2} 1 dx dy = \int_{y=0}^{y=1} \left[x \right]_{x=y^2}^{x=2y-y^2} dy \\ &= \int_{y=0}^{y=1} (2y - y^2) - y^2 dy = \left[y^2 - \frac{2}{3}y^3 \right]_{y=0}^{y=1} = \frac{1}{3}. \end{aligned}$$

Qu. 7. Sketch the region $R = \{(x, y, z) : 0 \leq z \leq 1 - |x| - |y|\}$ and evaluate the integral

$$\iiint_R (xy + z^2) dV.$$

Solution. The region is a pyramid with square base. To see this either consider cases $(x, y \geq 0; x \geq 0, y \leq 0; \text{etc})$ or note that the region is bounded by linear edges and planar faces and has vertices $(\pm 1, 0, 0)$, $(0, \pm 1, 0)$ and $(0, 0, 1)$.

Now let $f(x, y, z) = xy + z^2$ and let $I = \iiint_R f(x, y, z) dV$. Consider the four regions

$$\begin{aligned} R_1 &= \{(x, y, z) \in R : x \geq 0, y \geq 0\} & R_2 &= \{(x, y, z) \in R : x \leq 0, y \geq 0\} \\ R_3 &= \{(x, y, z) \in R : x \leq 0, y \leq 0\} & R_4 &= \{(x, y, z) \in R : x \geq 0, y \leq 0\}. \end{aligned}$$

Clearly I is the sum of the integrals of f over these four regions. Notice that $f(-x, -y, z) = f(x, y, z)$ and $f(x, -y, z) = f(-x, y, z)$, so

$$\iiint_{R_1} f dV = \iiint_{R_3} f dV \qquad \iiint_{R_2} f dV = \iiint_{R_4} f dV.$$

Hence $I = 2 \iiint_{R_1} f dV + 2 \iiint_{R_2} f dV$.

Now consider the first of these integrals. Note that x and y are both positive so that $1 - |x| - |y| = 1 - x - y$. We integrate first with respect to z , fixing x and y , so $0 \leq z \leq 1 - x - y$. We next integrate with respect to y fixing x , so $0 \leq y \leq 1 - x$,

since $0 \leq 1 - x - y$. Finally integrate with respect to x which varies between 0 and 1.

$$\begin{aligned}
\iiint_{R_1} f \, dV &= \iiint_{R_1} xy + z^2 \, dz \, dy \, dx = \int_{x=0}^{x=1} \int_{y=0}^{y=1-x} \int_{z=0}^{z=1-x-y} xy + z^2 \, dz \, dy \, dx \\
&= \int_{x=0}^{x=1} \int_{y=0}^{y=1-x} \left[xyz + \frac{1}{3} z^3 \right]_{z=0}^{z=1-x-y} dy \, dx \\
&= \int_{x=0}^{x=1} \int_{y=0}^{y=1-x} xy(1-x-y) + \frac{1}{3}(1-x-y)^3 \, dy \, dx \\
&= \int_{x=0}^{x=1} \int_{y=0}^{y=1-x} x(1-x)y - xy^2 + \frac{1}{3}(1-x-y)^3 \, dy \, dx \\
&= \int_{x=0}^{x=1} \left[\frac{1}{2}x(1-x)y^2 - \frac{1}{3}xy^3 - \frac{1}{12}(1-x-y)^4 \right]_{y=0}^{y=1-x} dx \\
&= \int_{x=0}^{x=1} \frac{1}{2}x(1-x)(1-x)^2 - \frac{1}{3}x(1-x)^3 + \frac{1}{12}(1-x)^4 \, dx \\
&= \int_{x=0}^{x=1} \frac{1}{6}x(1-x)^3 + \frac{1}{12}(1-x)^4 \, dx = \int_0^1 \frac{1}{6}(1-x)^3 - \frac{1}{12}(1-x)^4 \, dx \\
&= \left[-\frac{1}{24}(1-x)^4 + \frac{1}{60}(1-x)^5 \right]_{x=0}^{x=1} = \frac{1}{24} - \frac{1}{60}
\end{aligned}$$

Considering the second integral, $x \leq 0$ and $y \geq 0$, so that $1 - |x| - |y| = 1 + x - y$. Again integrating with respect to z first, $0 \leq 1 + x - y$. Fixing x , $0 \leq 1 + x - y$ so that $0 \leq y \leq 1 + x$. Finally $-1 \leq x \leq 0$. So

$$\begin{aligned}
&\iiint_{R_2} f \, dV \\
&= \int_{x=-1}^{x=0} \int_{y=0}^{y=1+x} \int_{z=0}^{z=1+x-y} xy + z^2 \, dz \, dy \, dx = \int_{x=-1}^{x=0} \int_{y=0}^{y=1+x} \left[xyz + \frac{1}{3} z^3 \right]_{z=0}^{z=1+x-y} dy \, dx \\
&= \int_{x=-1}^{x=0} \int_{y=0}^{y=1+x} xy(1+x-y) + \frac{1}{3}(1+x-y)^3 \, dy \, dx \\
&= \int_{x=-1}^{x=0} \left[\frac{1}{2}x(1+x)y^2 - \frac{1}{3}xy^3 - \frac{1}{12}(1+x-y)^4 \right]_{y=0}^{y=1+x} dx \\
&= \int_{x=-1}^{x=0} -\frac{1}{6}(1+x)^3 + \frac{1}{4}(1+x)^4 \, dx = \left[-\frac{1}{24}(1+x)^4 + \frac{1}{20}(1+x)^5 \right]_{x=-1}^{x=0} = -\frac{1}{24} + \frac{1}{20}.
\end{aligned}$$

Hence $I = 2\left(\frac{1}{24} - \frac{1}{60}\right) + 2\left(-\frac{1}{24} + \frac{1}{20}\right) = \frac{1}{15}$.
